

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Symmetric Algebraic Exponential Integral

Evaluate the definite integral:

$$\int_{-1}^1 \frac{(1 + \sqrt{|x|} - x)e^x + (\sqrt{|x|} - x)e^{-x}}{e^x + e^{-x}} dx$$

Solution: Let the target integral be denoted by I . First, we can simplify the numerator by expanding and grouping the terms strategically:

$$(1 + \sqrt{|x|} - x)e^x + (\sqrt{|x|} - x)e^{-x} = e^x + (\sqrt{|x|} - x)(e^x + e^{-x})$$

Dividing this by the denominator $(e^x + e^{-x})$, the integrand splits cleanly into two parts:

$$I = \int_{-1}^1 \left(\frac{e^x}{e^x + e^{-x}} + \sqrt{|x|} - x \right) dx$$

To evaluate this integral over the symmetric interval $[-1, 1]$, we utilize the property $\int_{-a}^a f(x) dx = \int_0^a (f(x) + f(-x)) dx$. Let $f(x) = \frac{e^x}{e^x + e^{-x}} + \sqrt{|x|} - x$. We compute $f(x) + f(-x)$ for $x \in [0, 1]$, where $|x| = x$:

$$f(x) = \frac{e^x}{e^x + e^{-x}} + \sqrt{x} - x$$

$$f(-x) = \frac{e^{-x}}{e^{-x} + e^x} + \sqrt{-x} - (-x) = \frac{e^{-x}}{e^x + e^{-x}} + \sqrt{x} + x$$

Adding these two functions together causes the x terms to cancel out, and the exponential fraction simplifies to 1:

$$f(x) + f(-x) = \frac{e^x + e^{-x}}{e^x + e^{-x}} + 2\sqrt{x} = 1 + 2\sqrt{x}$$

The integral I thus simplifies to a straightforward integral from 0 to 1:

$$I = \int_0^1 (1 + 2\sqrt{x}) dx$$

Applying the power rule for integration:

$$I = \left[x + \frac{4}{3}x^{3/2} \right]_0^1 = \left(1 + \frac{4}{3} \right) - 0 = \frac{7}{3}$$

1.2 Hard Integral

Double Error Function Integral

Evaluate the definite integral:

$$\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} \operatorname{erf}(x) \operatorname{erf}(y) dx dy$$

Solution: Notice that the double integral is completely separable into the product of two identical 1-dimensional integrals. Let J represent this 1D integral:

$$J = \int_0^\infty e^{-x^2} \operatorname{erf}(x) dx$$

Thus, the original double integral is simply J^2 . To evaluate J , we introduce a parameter a to construct a function $F(a)$:

$$F(a) = \int_0^\infty e^{-x^2} \operatorname{erf}(ax) dx$$

Our goal is to find $J = F(1)$. We differentiate $F(a)$ with respect to a under the integral sign. Recall that the derivative of the error function is $\frac{d}{dz} \operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} e^{-z^2}$:

$$F'(a) = \int_0^\infty e^{-x^2} \frac{\partial}{\partial a} \operatorname{erf}(ax) dx = \frac{2}{\sqrt{\pi}} \int_0^\infty x e^{-x^2} e^{-a^2 x^2} dx$$

$$F'(a) = \frac{2}{\sqrt{\pi}} \int_0^\infty x e^{-(1+a^2)x^2} dx$$

This resulting integral is straightforward to solve using the substitution $u = (1+a^2)x^2$, $du = 2(1+a^2)x dx$:

$$\int_0^\infty 2x e^{-(1+a^2)x^2} dx = \left[\frac{-1}{1+a^2} e^{-(1+a^2)x^2} \right]_0^\infty = \frac{1}{1+a^2}$$

Substituting this back gives $F'(a)$:

$$F'(a) = \frac{1}{\sqrt{\pi}(1+a^2)}$$

To find $F(1)$, we integrate $F'(a)$ from $a = 0$ to $a = 1$. Note that $F(0) = \int_0^\infty e^{-x^2} \operatorname{erf}(0) dx = 0$:

$$F(1) = \int_0^1 \frac{1}{\sqrt{\pi}(1+a^2)} da = \frac{1}{\sqrt{\pi}} \left[\tan^{-1}(a) \right]_0^1 = \frac{1}{\sqrt{\pi}} \left(\frac{\pi}{4} - 0 \right) = \frac{\sqrt{\pi}}{4}$$

Since $J = F(1) = \frac{\sqrt{\pi}}{4}$, our original double integral is J^2 :

$$\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} \operatorname{erf}(x) \operatorname{erf}(y) dx dy = \left(\frac{\sqrt{\pi}}{4} \right)^2 = \frac{\pi}{16}$$

Differentials

2.1 Medium Differential

Volterra Integral Equation

Solve the Volterra integral equation:

$$y(x) = \cos(x) + \int_0^x (x-t)y(t) dt$$

Given the initial conditions $y(0) = 1$ and $y'(0) = 0$, find the exact value of $y(\pi)$.

Solution: To solve this integral equation, we convert it into a differential equation by differentiating both sides with respect to x . Applying the Leibniz rule for differentiation under the integral sign:

$$y'(x) = -\sin(x) + \frac{d}{dx} \left(x \int_0^x y(t) dt - \int_0^x ty(t) dt \right)$$

$$y'(x) = -\sin(x) + \left(\int_0^x y(t) dt + xy(x) \right) - xy(x)$$

$$y'(x) = -\sin(x) + \int_0^x y(t) dt$$

We differentiate a second time with respect to x to eliminate the integral completely:

$$y''(x) = -\cos(x) + y(x)$$

Rearranging this yields a second-order linear non-homogeneous differential equation:

$$y'' - y = -\cos(x)$$

The characteristic equation for the homogeneous part $y'' - y = 0$ is $r^2 - 1 = 0$, yielding roots $r = \pm 1$. The complementary solution is:

$$y_c(x) = c_1 e^x + c_2 e^{-x}$$

For the particular solution $y_p(x)$, we guess the form $y_p(x) = A \cos(x) + B \sin(x)$. Differentiating gives $y_p''(x) = -A \cos(x) - B \sin(x)$. Substituting into the differential equation:

$$(-A \cos(x) - B \sin(x)) - (A \cos(x) + B \sin(x)) = -\cos(x)$$

$$-2A \cos(x) - 2B \sin(x) = -\cos(x)$$

By comparing coefficients, we get $-2A = -1 \implies A = 1/2$, and $-2B = 0 \implies B = 0$. The general solution is:

$$y(x) = c_1 e^x + c_2 e^{-x} + \frac{1}{2} \cos(x)$$

We apply the initial conditions to find the constants.

$$y(0) = 1 \implies c_1(1) + c_2(1) + \frac{1}{2}(1) = 1 \implies c_1 + c_2 = \frac{1}{2}$$

$$y'(x) = c_1e^x - c_2e^{-x} - \frac{1}{2}\sin(x)$$

$$y'(0) = 0 \implies c_1(1) - c_2(1) - 0 = 0 \implies c_1 = c_2$$

Solving this system gives $c_1 = c_2 = \frac{1}{4}$. The exact particular solution is:

$$y(x) = \frac{1}{4}e^x + \frac{1}{4}e^{-x} + \frac{1}{2}\cos(x) = \frac{1}{2}\cosh(x) + \frac{1}{2}\cos(x)$$

Finally, we evaluate the function at $x = \pi$:

$$y(\pi) = \frac{1}{2}\cosh(\pi) + \frac{1}{2}\cos(\pi) = \frac{\cosh(\pi) - 1}{2}$$

2.2 Hard Differential

Schrödinger Equation Power Series Limit

A quantum particle of mass m is governed by the one-dimensional time-independent Schrödinger equation. Under a suitable change of variables, the equation reduces to:

$$\frac{d^2\phi}{dz^2} = z\phi$$

where ϕ is the wave function and z is a dimensionless position. The general solution is $\phi(z) = c_1A(z) + c_2B(z)$, where $A(z)$ and $B(z)$ are power series centered at $z = 0$. Given $B(0) = 0$ and $B'(0) = 1$, evaluate:

$$\lim_{z \rightarrow 0} \frac{z^4}{B(z) - z}$$

Solution: This is the well-known Airy differential equation. Since $B(z)$ is analytic at $z = 0$, we can express it as a power series:

$$B(z) = \sum_{n=0}^{\infty} a_n z^n$$

Applying the initial conditions: $B(0) = 0 \implies a_0 = 0$ $B'(0) = 1 \implies a_1 = 1$ Now, substitute the series into the differential equation $B''(z) = zB(z)$:

$$\sum_{n=2}^{\infty} n(n-1)a_n z^{n-2} = z \sum_{n=0}^{\infty} a_n z^n = \sum_{n=0}^{\infty} a_n z^{n+1}$$

Shift the index on the left side by setting $k = n - 2$ ($n = k + 2$), and on the right side by setting $k = n + 1$ ($n = k - 1$):

$$\sum_{k=0}^{\infty} (k+2)(k+1)a_{k+2}z^k = \sum_{k=1}^{\infty} a_{k-1}z^k$$

For the equation to hold for all z , the coefficients of z^k on both sides must match. For $k = 0$ (the constant term):

$$(2)(1)a_2 = 0 \implies a_2 = 0$$

For $k \geq 1$, we equate the coefficients to obtain the recurrence relation:

$$(k+2)(k+1)a_{k+2} = a_{k-1} \implies a_{k+2} = \frac{a_{k-1}}{(k+2)(k+1)}$$

Using this relation, we find the subsequent coefficients: For $k = 1$: $a_3 = \frac{a_0}{3 \cdot 2} = 0$ (since $a_0 = 0$) For $k = 2$: $a_4 = \frac{a_1}{4 \cdot 3} = \frac{1}{12}$ (since $a_1 = 1$) For $k = 3$: $a_5 = \frac{a_2}{5 \cdot 4} = 0$ The power series expansion for $B(z)$ up to the $O(z^5)$ term is:

$$B(z) = z + \frac{1}{12}z^4 + O(z^7)$$

Substitute this series expansion into the limit expression:

$$\begin{aligned} \lim_{z \rightarrow 0} \frac{z^4}{B(z) - z} &= \lim_{z \rightarrow 0} \frac{z^4}{(z + \frac{1}{12}z^4 + O(z^7)) - z} \\ &= \lim_{z \rightarrow 0} \frac{z^4}{\frac{1}{12}z^4 + O(z^7)} \end{aligned}$$

Dividing the numerator and denominator by z^4 :

$$= \lim_{z \rightarrow 0} \frac{1}{\frac{1}{12} + O(z^3)} = \frac{1}{1/12} = 12$$

Derivatives

3.1 Medium Derivative

Implicit Exponential Derivative

Let $x^y = y^x$. Find $\frac{dy}{dx}\bigg|_{(2,4)}$.

Solution: To compute the derivative of this implicit relation where variables are present in both the bases and exponents, we apply logarithmic differentiation. We take the natural logarithm of both sides:

$$\ln(x^y) = \ln(y^x)$$

Using logarithmic properties, bring the exponents to the front:

$$y \ln(x) = x \ln(y)$$

Now, we differentiate implicitly with respect to x . We must apply the product rule to both sides:

$$\begin{aligned} \frac{d}{dx}[y \ln(x)] &= \frac{d}{dx}[x \ln(y)] \\ \frac{dy}{dx} \ln(x) + y \left(\frac{1}{x}\right) &= (1) \ln(y) + x \left(\frac{1}{y} \frac{dy}{dx}\right) \end{aligned}$$

Let $y' = \frac{dy}{dx}$. Rearrange the terms to group all expressions containing y' on the left side:

$$y' \ln(x) - \frac{x}{y} y' = \ln(y) - \frac{y}{x}$$

Factor out y' :

$$\begin{aligned} y' \left(\ln(x) - \frac{x}{y} \right) &= \ln(y) - \frac{y}{x} \\ y' &= \frac{\ln(y) - \frac{y}{x}}{\ln(x) - \frac{x}{y}} \end{aligned}$$

Now, substitute the coordinates of the given point $(x, y) = (2, 4)$ into the derivative expression:

$$y'|_{(2,4)} = \frac{\ln(4) - \frac{4}{2}}{\ln(2) - \frac{2}{4}} = \frac{\ln(4) - 2}{\ln(2) - \frac{1}{2}}$$

Since $\ln(4) = \ln(2^2) = 2 \ln(2)$, we can simplify the expression:

$$y'|_{(2,4)} = \frac{2 \ln(2) - 2}{\ln(2) - \frac{1}{2}}$$

Multiply the numerator and denominator by 2 to clear the fraction in the denominator:

$$\begin{aligned} y'|_{(2,4)} &= \frac{4 \ln(2) - 4}{2 \ln(2) - 1} \\ y'|_{(2,4)} &= \frac{4(\ln(2) - 1)}{2 \ln(2) - 1} \end{aligned}$$

3.2 Hard Derivative

High Order Maclaurin Derivative

Let $f(x) = \frac{1}{1+x+x^2}$. Find $\frac{f^{(2026)}(0)}{2026!}$.

Solution: The expression $\frac{f^{(2026)}(0)}{2026!}$ is precisely the definition of the coefficient of x^{2026} in the Maclaurin series expansion of $f(x)$. Our task is to find the Maclaurin series for this rational function. We can manipulate the function to reveal a geometric series structure. By multiplying the numerator and denominator by $(1-x)$, we leverage the difference of cubes factorization $1-x^3 = (1-x)(1+x+x^2)$:

$$f(x) = \frac{1}{1+x+x^2} \cdot \frac{1-x}{1-x} = \frac{1-x}{1-x^3}$$

This expression can be decomposed into a product of $(1-x)$ and a standard geometric series with common ratio x^3 :

$$f(x) = (1-x) \sum_{n=0}^{\infty} (x^3)^n = (1-x) \sum_{n=0}^{\infty} x^{3n}$$

Expanding this product:

$$f(x) = \sum_{n=0}^{\infty} x^{3n} - \sum_{n=0}^{\infty} x^{3n+1}$$

This shows that the Maclaurin series consists of terms with powers that are multiples of 3, and terms with powers that are one more than a multiple of 3. Specifically: - If the power $k = 3n$, the coefficient is 1. - If the power $k = 3n + 1$, the coefficient is -1 . - If the power $k = 3n + 2$, the coefficient is 0. We need the coefficient of x^{2026} . Let's divide 2026 by 3 to determine its form:

$$2026 = 3 \times 675 + 1$$

Since 2026 is of the form $3n + 1$, the corresponding coefficient in the Maclaurin series is exactly -1 . Therefore, by Taylor's theorem:

$$\frac{f^{(2026)}(0)}{2026!} = -1$$

Physics & Engineering

4.1 Electromagnetism

Spherical Capacitor

A spherical capacitor consists of two concentric conducting shells. The inner shell has a radius of $a = 0.1$ m and the outer grounded shell has a radius of $b = 0.3$ m. Calculate the capacitance of this system in picofarads (pF). Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$.

Solution: The capacitance C of a spherical capacitor formed by two concentric shells of radii a and b (where $b > a$) is given by the formula:

$$C = \frac{4\pi\epsilon_0 ab}{b - a}$$

We are given the Coulomb's constant $k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$. Therefore, $4\pi\epsilon_0 = \frac{1}{9 \times 10^9} \text{ C}^2/(\text{N} \cdot \text{m}^2)$. Substitute the radii $a = 0.1$ m and $b = 0.3$ m into the formula:

$$C = \left(\frac{1}{9 \times 10^9} \right) \frac{(0.1)(0.3)}{0.3 - 0.1}$$

$$C = \left(\frac{1}{9 \times 10^9} \right) \frac{0.03}{0.2} = \left(\frac{1}{9 \times 10^9} \right) (0.15)$$

$$C = \frac{15 \times 10^{-2}}{9 \times 10^9} = \frac{15}{9} \times 10^{-11} = \frac{5}{3} \times 10^{-11} \text{ F}$$

To convert Farads (F) to picofarads (pF), we multiply by 10^{12} :

$$C \text{ (in pF)} = \left(\frac{5}{3} \times 10^{-11} \right) \times 10^{12} = \frac{50}{3} \text{ pF} \approx 16.67 \text{ pF}$$

4.2 Quantum Mechanics

Coherent State Expectation Value

A particle in the harmonic oscillator potential is in a coherent state $|\phi\rangle$ such that $a|\phi\rangle = \alpha|\phi\rangle$. Find the expectation value of the quantum number operator N if $\alpha = \frac{3}{2} + \frac{i}{2}$.

Solution: The quantum number operator (or number operator) for a harmonic oscillator is defined as $N = a^\dagger a$, where a is the annihilation operator and $a^\dagger$ is the creation operator. The expectation value of N in the coherent state $|\phi\rangle$ is given by:

$$\langle N \rangle = \langle \phi | N | \phi \rangle = \langle \phi | a^\dagger a | \phi \rangle$$

We are given that $|\phi\rangle$ is an eigenstate of the annihilation operator a with eigenvalue α :

$$a|\phi\rangle = \alpha|\phi\rangle$$

Taking the Hermitian conjugate of this eigenvalue equation yields:

$$\langle\phi|a^\dagger = \alpha^*\langle\phi|$$

Substituting these relations into the expectation value expression:

$$\langle N \rangle = (\langle\phi|a^\dagger)(a|\phi\rangle) = (\alpha^*\langle\phi|)(\alpha|\phi\rangle) = \alpha^*\alpha\langle\phi|\phi\rangle$$

Assuming the coherent state $|\phi\rangle$ is normalized ($\langle\phi|\phi\rangle = 1$):

$$\langle N \rangle = \alpha^*\alpha = |\alpha|^2$$

We are given the complex eigenvalue $\alpha = \frac{3}{2} + \frac{i}{2}$. The magnitude squared of this complex number is the sum of the squares of its real and imaginary parts:

$$|\alpha|^2 = \left(\frac{3}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{9}{4} + \frac{1}{4} = \frac{10}{4} = 2.5$$

Therefore, the expectation value of the quantum number operator is 2.5.

4.3 Mechanics

Orbital Velocity at Periapsis

A planet moves around a star of mass $M = 1.00 \times 10^{30}$ kg in an elliptical orbit. The minimum distance is $r_{min} = 2.00 \times 10^{11}$ m and the maximum distance is $r_{max} = 8.00 \times 10^{11}$ m. The speed at the point of closest approach is 10^α m s⁻¹. Find the value of α . Take $G = 6.67 \times 10^{-11}$ N m² kg⁻².

Solution: The semi-major axis a of the elliptical orbit is the average of the minimum (periapsis) and maximum (apoapsis) distances:

$$a = \frac{r_{min} + r_{max}}{2} = \frac{2.00 \times 10^{11} + 8.00 \times 10^{11}}{2} = 5.00 \times 10^{11} \text{ m}$$

The velocity v of the planet at any distance r from the star is given by the vis-viva equation:

$$v^2 = GM \left(\frac{2}{r} - \frac{1}{a} \right)$$

At the point of closest approach, $r = r_{min} = 2.00 \times 10^{11}$ m. Substituting the known values:

$$v^2 = (6.67 \times 10^{-11})(1.00 \times 10^{30}) \left(\frac{2}{2.00 \times 10^{11}} - \frac{1}{5.00 \times 10^{11}} \right)$$

First, compute the standard gravitational parameter $\mu = GM$:

$$\mu = 6.67 \times 10^{19} \text{ m}^3 \text{ s}^{-2}$$

Next, evaluate the terms inside the parenthesis:

$$\frac{2}{2 \times 10^{11}} - \frac{1}{5 \times 10^{11}} = \frac{1}{10^{11}} - \frac{0.2}{10^{11}} = \frac{0.8}{10^{11}} = 8 \times 10^{-12} \text{ m}^{-1}$$

Now find the squared velocity:

$$v^2 = (6.67 \times 10^{19}) \times (8 \times 10^{-12}) = 53.36 \times 10^7 = 5.336 \times 10^8 \text{ m}^2\text{s}^{-2}$$

The speed v is the square root of this value:

$$v = \sqrt{5.336 \times 10^8} \text{ m/s}$$

We are given that the speed is 10^α . To find α , we take the base-10 logarithm of v :

$$10^\alpha = \sqrt{5.336 \times 10^8} \implies \alpha = \log_{10} \left((5.336 \times 10^8)^{1/2} \right)$$

$$\alpha = \frac{1}{2} \log_{10}(5.336 \times 10^8) = \frac{1}{2} (\log_{10}(5.336) + 8)$$

$$\alpha = \frac{1}{2} (0.7272 + 8) = \frac{8.7272}{2} \approx 4.3636$$

Therefore, $\alpha \approx 4.364$.